diamond

Doc Type	Cause & Effect Matrix
Title	PSS B10E HeXi Cause & Effect Matrix
Doc Number	TDI-PSS-B10E-C&E Version 4.0

Notes:
Prepared in conformity with:
IEC 62881 Cause & Effect Matrix
IEC 62708 Document kinds for Electrical and Instrumentation Projects in the Process Industry

Rev. History

Date	Rev.	Description	Originator	Review/Checker	Approver
01-Dec-25	1.0	First Draft	JF	MW	
17-Feb-26	2.0	Updated following comments & peer review	JF	JA	AA
16-Mar-26	3.0	Alarm Points updated	MW		
14/04/26	4.0	Additional 3 contactors (Con 6,7,8) for the gate. Split of safety functions for gate and door. Inclusion of 200ms delay (Con 6,7,8) . Additional contactor for the RF to be removed when the gate or door has a demand. When a Beam off Button (BoB) is depressed the safety function will remove contactors (01,02,03,06,07,08,09,10) ensuring the hazard is removed in two ways. Remove the SF6 sensors, as hazard will be detected by 02 detectors (03,04) SIF 09 Radiation Monitor(s) Updated to include RF Contactor 09.	MW	AA	JA

References:

TDI-PSS-POL-0022 PSS Naming Conventions
Defines Tag names as used in Diamond Light Source Personnel Safety Systems

TDI-PSS-SRS-0002 Safety Requirement Specification for B10E

Dwg 1224211 B10E SIF Logic

Dwg 1212854 B10E PSS Layout

BS EN IEC 62881:2018 Cause and Effect Matrix
Layout, mandatory and optional attributes of causes, effects and relations used in this document have been designed according to requirements of clause 5 (Design of C&E matrices).
Attributes of relations will use Alternative 2 to define simple description of triggered action as per clause 5.4
- CL = Closed
- OP = Open
- On/Off

Relations are shown in the activated (1) condition.
Unless stated within description, effects are de-energise to trip (0)

Footnotes:

Search function is sequential. Order of sequence transitions noted in relation field actions by subscript Tx
Further details on search sequence provided in logic drawing XXXXXXX

[a] 20s delay applies to SIF-03 only. SIF 08 Beam Off Button (BOB) activated General and Service Access Door Unlock (GADL & SADL)must be instant.
[b] SIF-03 prevents access to hazardous area by keeping General and Service Access Doors Locked (GADL & SADL) while radiation hazard is present whilst also allowing access by Unlocking doors and indicating 'OPEN' status on Annunciator (ANNOP) when radiation hazard is absent.
[c] 180s delay after General and Service Access Doors Locked (GADL & SADL) and SIF-06 satisfied before 'BEAM ON' Indicator (BONI) and 'BEAM ON' Annunciator (ANNBON)status.
[d] 180s delay after Area Search Button 05 (ASB05) pressed in search sequence, SIF-06 only.
[e] 180s delay to General Access Door Locked (GADL) does not apply to SIF-03.
[f] Search timer of 180s commences on sequence transition T1. 'RESTRICTED' Annunciator (ANNRES), Speaker Program 1 (SP01) remain active until SIF-06 is satisfied or timer elapsed. Service Access Door remains Locked (SADL) if SIF-06 satisfied.
[g] General Access Door and Gate Closed (GADC) before Area Search Button 05 (ASB05) activated.
[h] Group for voting arrangement: Letters assigned to Sensor groups, Numbers assigned to Final Element groups. Details on voting between groups given in SRS/SIF document TDI-PSS-SRS-0002
[i] Light curtain consists of two components, receiver LCRx & transmitter LCTx. LCRx is sensor listed as cause. LCRx activation/interrupt of beam = (0). Detailed function of light curtain:
- Registers 1 activation (Single Pass-Through) (0) Prior to search sequence T3
- Monitored T3 - T6, to result in trip if activated (0)
- Monitoring ceases after T6
[j] Beam Off Buttons (BOB), Blue Lights (BLUEL), Annunciator Functions (ANNUN, ANNRES, ANNSTD, ANNBON), Beam On Indicator (BONI) & Beam Off Button Indicators (BOBI) are all dual channel devices.
[k] Electron Source Enable Key has 2 locations. First location B10E-PS-KEY-02 must be On for search start T1. Key is retained in On condition for duration of search. Once Transition T7 is achieved, Key is released and can be turned Off to travel to new location B10E-PS-KEY-03. Removal of key at this time only does not break search logic. Removal of key prior to transition T7 must result in search failure.
[l] Annunciator 'OPEN' status remains off until current and voltage transmitters read 0 and a time delay of 12 seconds has passed and open/reset function is pressed. See B10E-SIF-03 sheet 4 of drawing 1224211

Contactors 01,

Rev	Simplified Signal Tag	Description	Device Tag	EPICS/DCS Tag	Reference Doc(s)	Detailed Function	Fail Safe Position	Voting (to trip)	Group[h]	Timing	SIL	SIF(s)
4.0	CON10	Power Contactor 10	B10E-PS-CON-10	B10E-PS-CON-10	Dwg 1224211 Sheet 9,12	Isolation of RF Source (Radiation Hazard)	De-Energised/Open Circuit	1oo1	15	200ms	2	B10E-SIF-10
4.0	CON09	Power Contactor 09	B10E-PS-CON-09	B10E-PS-CON-09	Dwg 1224211 Sheet 2,3,9	Isolation of RF Source (Radiation Hazard)	De-Energised/Open Circuit	1oo1	14	N/a	2	B10E-SIF-10
4.0	CON08	Power Contactor 08	B10E-PS-CON-08	B10E-PS-CON-08	Dwg 1224211 Sheet 9,12	Isolation of Electron Source (Radiation Hazard)	De-Energised/Open Circuit	1oo3	13	200ms	2	B10E-SIF-13
4.0	CON07	Power Contactor 07	B10E-PS-CON-07	B10E-PS-CON-07	Dwg 1224211 Sheet 9,12	Isolation of Electron Source (Radiation Hazard)	De-Energised/Open Circuit	1oo3	13	200ms	2	B10E-SIF-13
4.0	CON06	Power Contactor 06	B10E-PS-CON-06	B10E-PS-CON-06	Dwg 1224211 Sheet 9,12	Isolation of Electron Source (Radiation Hazard)	De-Energised/Open Circuit	1oo3	13	200ms	2	B10E-SIF-13
2.0	IND03	Traffic Light/ Alarm Indicator	B10E-PS-IND-03	B10E-PS-IND-03	Dwg 1224211 Sheet 11	O2 Depletion/Toxic Gas Status, Visual Indicator (Red/Green)	De-Energised/Off	N/a	12	N/a		B10E-SIF-11 & 12
2.0	IND02	Traffic Light/ Alarm Indicator	B10E-PS-IND-02	B10E-PS-IND-02	Dwg 1224211 Sheet 11	O2 Depletion/Toxic Gas Status, Visual Indicator (Red/Green)	De-Energised/Off	N/a	12	N/a		B10E-SIF-11 & 12
2.0	IND01	Traffic Light/ Alarm Indicator	B10E-PS-IND-01	B10E-PS-IND-01	Dwg 1224211 Sheet 11	O2 Depletion/Toxic Gas Status, Visual Indicator (Red/Green)	De-Energised/Off	N/a	12	N/a		B10E-SIF-11 & 12
2.0	BOBI05[j]	Beam Off Button Indicator 04	B10E-PS-BOBI-04	B10E-PS-BOBI-04	Dwg 1224211 Sheet 7	Deterrant Message Display 'PRESS BEAM OFF'	Inhibit B10E-SIF-04 & 05	N/a	11	180s[d]		B10E-SIF-06
2.0	BOBI04[j]	Beam Off Button Indicator 04	B10E-PS-BOBI-04	B10E-PS-BOBI-04	Dwg 1224211 Sheet 7	Deterrant Message Display 'PRESS BEAM OFF'	Inhibit B10E-SIF-04 & 05	N/a	11	180s[d]		B10E-SIF-06
2.0	BOBI03[j]	Beam Off Button Indicator 03	B10E-PS-BOBI-03	B10E-PS-BOBI-03	Dwg 1224211 Sheet 7	Deterrant Message Display 'PRESS BEAM OFF'	Inhibit B10E-SIF-04 & 05	N/a	11	180s[d]		B10E-SIF-06
2.0	BOBI02[j]	Beam Off Button Indicator 02	B10E-PS-BOBI-02	B10E-PS-BOBI-02	Dwg 1224211 Sheet 7	Deterrant Message Display 'PRESS BEAM OFF'	Inhibit B10E-SIF-04 & 05	N/a	11	180s[d]		B10E-SIF-06
2.0	BOBI01[j]	Beam Off Button Indicator 01	B10E-PS-BOBI-01	B10E-PS-BOBI-01	Dwg 1224211 Sheet 7	Deterrant Message Display 'PRESS BEAM OFF'	Inhibit B10E-SIF-04 & 05	N/a	11	180s[d]		B10E-SIF-06
2.0	BONI[j]	Beam On Indicator	B10E-PS-BONI-01	B10E-PS-BONI-01	Dwg 1224211 Sheet 8	Deterrant Message Display 'BEAM ON'	Inhibit B10E-SIF-04 & 05	N/a	6	180s[c]		B10E-SIF-07
2.0	SDVLN2	Liquid Nitrogen Shutoff Valve	B10E-PS-SDVLN2-01	B10E-PS-SDVLN2-01	Dwg 1224211 Sheet 11	Isolation of LN2 Hazard	De-energised/Valve Closed	1oo1	N/a	N/a		B10E-SIF-11
2.0	BLUEL04[j]	Blue Light 4	B10E-PS-BLUEL-04	B10E-PS-BLUEL-04	Dwg 1224211 Sheet 7	Deterrant Light Visual	Inhibit B10E-SIF-04 & 05	N/a	10	180s[d]		B10E-SIF-06
2.0	BLUEL03[j]	Blue Light 3	B10E-PS-BLUEL-03	B10E-PS-BLUEL-03	Dwg 1224211 Sheet 7	Deterrant Light Visual	Inhibit B10E-SIF-04 & 05	N/a	10	180s[d]		B10E-SIF-06
2.0	BLUEL02[j]	Blue Light 2	B10E-PS-BLUEL-02	B10E-PS-BLUEL-02	Dwg 1224211 Sheet 7	Deterrant Light Visual	Inhibit B10E-SIF-04 & 05	N/a	10	180s[d]		B10E-SIF-06
2.0	BLUEL01[j]	Blue Light 1	B10E-PS-BLUEL-01	B10E-PS-BLUEL-01	Dwg 1224211 Sheet 7	Deterrant Light Visual	Inhibit B10E-SIF-04 & 05	N/a	10	180s[d]		B10E-SIF-06
2.0	SPR03	Speaker Program 3	B10E-PS-SP-02	B10E-PS-SP-02	Dwg 1224211 Sheet 11	Alarm Message Audio & Visual	De-Energised/Off	N/a	9	N/a		B10E-SIF-11 & 12
2.0	SPR03	Speaker Program 3	B10E-PS-SP-01	B10E-PS-SP-01	Dwg 1224211 Sheet 11	Alarm Message Audio & Visual	De-Energised/Off	N/a	9	N/a		B10E-SIF-11 & 12
2.0	SPR02	Speaker Program 2	B10E-PS-SP-02	B10E-PS-SP-02	Dwg 1224211 Sheet 7	Deterrant Message Audio & Visual	Inhibit B10E-SIF-04 & 05	N/a	8	180s[d]		B10E-SIF-06
2.0	SPR02	Speaker Program 2	B10E-PS-SP-01	B10E-PS-SP-01	Dwg 1224211 Sheet 7	Deterrant Message Audio & Visual	Inhibit B10E-SIF-04 & 05	N/a	8	180s[d]		B10E-SIF-06
2.0	SP01	Speaker Program 1	B10E-PS-SP-02	B10E-PS-SP-02	Dwg 1224211 Sheet 6	Cautionary Message Audio & Visual	Inhibit B10E-SIF-04 & 05	N/a	7	180s[f]		B10E-SIF-05
2.0	SP01	Speaker Program 1	B10E-PS-SP-01	B10E-PS-SP-01	Dwg 1224211 Sheet 6	Cautionary Message Audio & Visual	Inhibit B10E-SIF-04 & 05	N/a	7	180s[f]		B10E-SIF-05
2.0	ANNBON[j]	Annunciator Beam On	B10E-PS-ANNBON-01	B10E-PS-ANNBON	Dwg 1224211 Sheet 8	Deterrant Message Display 'BEAM ON'	Inhibit B10E-SIF-04 & 05	N/a	6	180s[c]		B10E-SIF-07
2.0	ANNSTD[j]	Annunciator Standby	B10E-PS-ANNUN-01	B10E-PS-ANNSTD	Dwg 1224211 Sheet 7	Deterrant Message Display 'STANDBY'	Inhibit B10E-SIF-04 & 05	N/a	N/a	N/a		B10E-SIF-06
2.0	ANNRES[j]	Annunciator Restricted	B10E-PS-ANNUN-01	B10E-PS-ANNRES	Dwg 1224211 Sheet 6	Cautionary Message Display 'RESTRICTED'	Inhibit B10E-SIF-04 & 05	N/a	N/a	180s[f]		B10E-SIF-05
2.0	SADL02	Service Access Door Lock 02	B10E-PS-DS-08	B10E-PS-ANNOPN	Dwg 1224211 Sheet 5	Status Message Display 'OPEN'	Inhibit B10E-SIF-04 & 05	N/a	N/a	N/a		B10E-SIF-03
2.0	SADL01	Service Access Door Lock 01	B10E-PS-DS-07	B10E-PS-SOL-06	Dwg 1224211 Sheets 4, 6 & 9	Lock of hazrdous Area Door	De-Energised/Unlocked	1oo2	5	180s[l]		B10E-SIF-03[a],[b],[e] & 08
2.0	GADL04	General Access Gate Lock 04	B10E-PS-DS-05	B10E-PS-SOL-05	Dwg 1224211 Sheets 4, 6 & 9	Lock of hazrdous Area Door	De-Energised/Unlocked	1oo2	4	180s[e]		B10E-SIF-03[a],[b],[e] & 08
2.0	GADL03	General Access Gate Lock 03	B10E-PS-DS-04	B10E-PS-SOL-04	Dwg 1224211 Sheets 4, 7 & 9	Lock of hazrdous Area Door	De-Energised/Unlocked	1oo2	4	180s[e]		B10E-SIF-03[a],[b],[e] & 08
2.0	GADL02	General Access Door Lock 02	B10E-PS-DS-02	B10E-PS-SOL-03	Dwg 1224211 Sheets 4, 7 & 9	Lock of hazrdous Area Door	De-Energised/Unlocked	1oo2	3	180s[e]		B10E-SIF-03[a],[b],[e] & 08
2.0	GADL01	General Access Door Lock 01	B10E-PS-DS-01	B10E-PS-SOL-01	Dwg 1224211 Sheets 4, 7 & 9	Lock of hazrdous Area Door	De-Energised/Unlocked	1oo2	3	180s[e]		B10E-SIF-03[a],[b],[e] & 08
2.0	CON05	Power Contactor 05	B10E-PS-CON-05	B10E-PS-CON-05	Dwg 1224211 Sheet 10	Isolation of Electron Source (Radiation Hazard)	De-Energised/Open Circuit	1oo2	2	N/a		B10E-SIF-10
2.0	CON04	Power Contactor 04	B10E-PS-CON-04	B10E-PS-CON-04	Dwg 1224211 Sheet 10	Isolation of Electron Source (Radiation Hazard)	De-Energised/Open Circuit	1oo2	2	N/a		B10E-SIF-10
4.0	CON03	Power Contactor 03	B10E-PS-CON-03	B10E-PS-CON-03	Dwg 1224211 Sheets 2, 3, 9 & 10	Isolation of Electron Source (Radiation Hazard)	De-Energised/Open Circuit	1oo3	1	N/a	2	B10E-SIF-01, 02, 08 & 09
4.0	CON02	Power Contactor 02	B10E-PS-CON-02	B10E-PS-CON-02	Dwg 1224211 Sheet 2, 3, 9 & 10	Isolation of Electron Source (Radiation Hazard)	De-Energised/Open Circuit	1oo3	1	N/a	2	B10E-SIF-01, 02, 08 & 09
4.0	CON01	Power Contactor 01	B10E-PS-CON-01	B10E-PS-CON-01	Dwg 1224211 Sheet 2, 3, 9 & 10	Isolation of Electron Source (Radiation Hazard)	De-Energised/Open Circuit	1oo3	1	N/a	2	B10E-SIF-01, 02, 08 & 09
2.0	SADL02	Service Access Door Lock 02	B10E-PS-DS-08	B10E-PS-ANNOPN	Dwg 1224211 Sheet 5	Status Message Display 'OPEN'	Inhibit B10E-SIF-04 & 05	1oo1	N/a	N/a		B10E-SIF-03
2.0	SADL01	Service Access Door Lock 01	B10E-PS-DS-07	B10E-PS-SOL-06	Dwg 1224211 Sheets 4, 6 & 9	Lock of hazrdous Area Door	De-Energised/Unlocked	1oo2	5	180s[l]		B10E-SIF-03[a],[b],[e] & 08
2.0	GADL04	General Access Gate Lock 04	B10E-PS-DS-05	B10E-PS-SOL-05	Dwg 1224211 Sheets 4, 6 & 9	Lock of hazrdous Area Door	De-Energised/Unlocked	1oo2	4	180s[e]		B10E-SIF-03[a],[b],[e] & 08
2.0	GADL03	General Access Gate Lock 03	B10E-PS-DS-04	B10E-PS-SOL-04	Dwg 1224211 Sheets 4, 7 & 9	Lock of hazrdous Area Door	De-Energised/Unlocked	1oo2	4	180s[e]		B10E-SIF-03[a],[b],[e] & 08
2.0	GADL02	General Access Door Lock 02	B10E-PS-DS-02	B10E-PS-SOL-03	Dwg 1224211 Sheets 4, 7 & 9	Lock of hazrdous Area Door	De-Energised/Unlocked	1oo2	3	180s[e]		B10E-SIF-03[a],[b],[e] & 08
2.0	GADL01	General Access Door Lock 01	B10E-PS-DS-01	B10E-PS-SOL-01	Dwg 1224211 Sheets 4, 7 & 9	Lock of hazrdous Area Door	De-Energised/Unlocked	1oo2	3	180s[e]		B10E-SIF-03[a],[b],[e] & 08
2.0	CON05	Power Contactor 05	B10E-PS-CON-05	B10E-PS-CON-05	Dwg 1224211 Sheet 10	Isolation of Electron Source (Radiation Hazard)	De-Energised/Open Circuit	1oo2	2	N/a		B10E-SIF-10
2.0	CON04	Power Contactor 04	B10E-PS-CON-04	B10E-PS-CON-04	Dwg 1224211 Sheet 10	Isolation of Electron Source (Radiation Hazard)	De-Energised/Open Circuit	1oo2	2	N/a		B10E-SIF-10
4.0	CON03	Power Contactor 03	B10E-PS-CON-03	B10E-PS-CON-03	Dwg 1224211 Sheets 2, 3, 9 & 10	Isolation of Electron Source (Radiation Hazard)	De-Energised/Open Circuit	1oo3	1	N/a	2	B10E-SIF-01, 02, 08 & 09
4.0	CON02	Power Contactor 02	B10E-PS-CON-02	B10E-PS-CON-02	Dwg 1224211 Sheet 2, 3, 9 & 10	Isolation of Electron Source (Radiation Hazard)	De-Energised/Open Circuit	1oo3	1	N/a	2	B10E-SIF-01, 02, 08 & 09
4.0	CON01	Power Contactor 01	B10E-PS-CON-01	B10E-PS-CON-01	Dwg 1224211 Sheet 2, 3, 9 & 10	Isolation of Electron Source (Radiation Hazard)	De-Energised/Open Circuit	1oo3	1	N/a	2	B10E-SIF-01, 02, 08 & 09

Effect

CAUSE

Rev	Simplified Signal Tag	Description	Device Tag	EPICS/DCS Tag	Reference Doc(s)	Activation Condition	Alarm	Trip Setting	Voting (to trip)	Group[h]	Sequential Order	SIL	SIF(s)
4.0	GADC01	General Access Door Closed 01	B10E-PS-DS-01	B10E-PS-GADC-01	Dwg 1224211 Sheets 2 & 7	Digital	None	0	2oo3	A	T7[g]	2	B10E-SIF-01 & 06
4.0	GADC02	General Access Door Closed 02	B10E-PS-DS-02	B10E-PS-GADC-02	Dwg 1224211 Sheets 2 & 7	Digital	None	0	2oo3	A	T7[g]	2	B10E-SIF-01 & 06
4.0	GADC03	General Access Door Closed 03	B10E-PS-DS-03	B10E-PS-GADC-03	Dwg 1224211 Sheets 2 & 7	Digital	None	0	2oo3	A	T7[g]	2	B10E-SIF-01 & 06
4.0	GADC04	General Access Gate Closed 04	B10E-PS-DS-04	B10E-PS-GADC-04	Dwg 1224211 Sheets 2 & 7	Digital	None	0	2oo3	B	T7[g]	2	B10E-SIF-01 & 06
4.0	GADC05	General Access Gate Closed 05	B10E-PS-DS-05	B10E-PS-GADC-05	Dwg 1224211 Sheets 2 & 7	Digital	None	0	2oo3	B	T7[g]	2	B10E-SIF-01 & 06
4.0	GADC06	General Access Gate Closed 06	B10E-PS-DS-06	B10E-PS-GADC-06	Dwg 1224211 Sheets 2 & 7	Digital	None	0	2oo3	B	T7[g]	2	B10E-SIF-01 & 06
3.0	SADC01	Service Access Door Closed 01	B10E-PS-DS-07	B10E-PS-SADC-01	Dwg 1224211 Sheets 3 & 6	Digital	None	0	2oo3	C	T1		B10E-SIF-02 & 05
3.0	SADC02	Service Access Door Closed 02	B10E-PS-DS-08	B10E-PS-SADC-02	Dwg 1224211 Sheets 3 & 6	Digital	None	0	2oo3	C	T1		B10E-SIF-02 & 05
3.0	SADC03	Service Access Door Closed 03	B10E-PS-DS-09	B10E-PS-SADC-03	Dwg 1224211 Sheets 3 & 6	Digital	None	0	2oo3	C	T1		B10E-SIF-02 & 05
3.0	SADC04	Service Access Door Closed 04	B10E-PS-DS-10	B10E-PS-SADC-04	Dwg 1224211 Sheets 3 & 6	Digital	None	0	1oo2	D	T1		B10E-SIF-02 & 05

| SIF(s) | CL | CL | CL

3.0	SADC05	Service Access Door Closed 05	B10E-PS-DS-11	B10E-PS-SADC-05	Dwg 1224211 Sheets 3 & 6	Digital	None	0	1oo2	D	T1		B10E-SIF-02 & 05			CL	CL	CL						CL_{T2}	CL_{T1}		On_{T1}			On_{T1}	On_{T1}																					
2.0	IT01	Current Transmitter 01	B10E-PS-IT-01	B10E-PS-IT-01	Dwg 1224211 Sheet 4	Analogue 4-20mA			2oo3	E	N/a		B10E-SIF-03[a]								CL^b	CL^b	CL^b	CL^b	CL^b	CL^b	Off^f																									
2.0	IT02	Current Transmitter 02	B10E-PS-IT-02	B10E-PS-IT-02	Dwg 1224211 Sheet 4	Analogue 4-20mA			2oo3	E	N/a		B10E-SIF-03[a]								CL^b	CL^b	CL^b	CL^b	CL^b	CL^b	Off^f																									
2.0	IT03	Current Transmitter 03	B10E-PS-IT-03	B10E-PS-IT-03	Dwg 1224211 Sheet 4	Analogue 4-20mA			2oo3	E	N/a		B10E-SIF-03[a]								CL^b	CL^b	CL^b	CL^b	CL^b	CL^b	Off^f																									
2.0	VT01	Voltage Transmitter 01	B10E-PS-VT-01	B10E-PS-VT-01	Dwg 1224211 Sheet 4	Analogue 4-20mA			2oo3	F	N/a		B10E-SIF-03[a]								CL^b	CL^b	CL^b	CL^b	CL^b	CL^b	Off^f																									
2.0	VT02	Voltage Transmitter 02	B10E-PS-VT-02	B10E-PS-VT-02	Dwg 1224211 Sheet 4	Analogue 4-20mA			2oo3	F	N/a		B10E-SIF-03[a]								CL^b	CL^b	CL^b	CL^b	CL^b	CL^b	Off^f																									
2.0	VT03	Voltage Transmitter 03	B10E-PS-VT-03	B10E-PS-VT-03	Dwg 1224211 Sheet 4	Analogue 4-20mA			2oo3	F	N/a		B10E-SIF-03[a]								CL^b	CL^b	CL^b	CL^b	CL^b	CL^b	Off^f																									
2.0	OXMON01	Oxygen Monitor 01	B10E-PS-OXMON-01	B10E-PS-OXMON-01	Dwg 1224211 Sheet 11	Analogue 4-20mA			1oo4	H	N/a		B10E-SIF-11																			On	On				OP						On	On	On						
2.0	OXMON02	Oxygen Monitor 02	B10E-PS-OXMON-02	B10E-PS-OXMON-02	Dwg 1224211 Sheet 11	Analogue 4-20mA			1oo4	H	N/a		B10E-SIF-11																			On	On				OP						On	On	On						
2.0	OXMON03	Oxygen Monitor 03	B10E-PS-OXMON-03	B10E-PS-OXMON-03	Dwg 1224211 Sheet 11	Analogue 4-20mA			1oo4	H	N/a		B10E-SIF-11																			On	On				OP						On	On	On						
2.0	OXMON04	Oxygen Monitor 04	B10E-PS-OXMON-04	B10E-PS-OXMON-04	Dwg 1224211 Sheet 11	Analogue 4-20mA			1oo4	H	N/a		B10E-SIF-11																			On	On				OP						On	On	On						
2.0	BOB01[j]	Beam Off Button 01	B10E-PS-BOB-01	B10E-PS-BOB-01	Dwg 1224211 Sheet 9	Digital		0	1oo7	J	N/a		B10E-SIF-08			CL	CL	CL			CL	CL	CL	CL	CL	CL																									
2.0	BOB02[j]	Beam Off Button 02	B10E-PS-BOB-02	B10E-PS-BOB-02	Dwg 1224211 Sheet 9	Digital		0	1oo7	J	N/a		B10E-SIF-08			CL	CL	CL			CL	CL	CL	CL	CL	CL																									
2.0	BOB03[j]	Beam Off Button 03	B10E-PS-BOB-03	B10E-PS-BOB-03	Dwg 1224211 Sheet 9	Digital		0	1oo7	J	N/a		B10E-SIF-08			CL	CL	CL			CL	CL	CL	CL	CL	CL																									
2.0	BOB04[j]	Beam Off Button 04	B10E-PS-BOB-04	B10E-PS-BOB-04	Dwg 1224211 Sheet 9	Digital		0	1oo7	J	N/a		B10E-SIF-08			CL	CL	CL			CL	CL	CL	CL	CL	CL																									
2.0	BOB05[j]	Beam Off Button 05	B10E-PS-BOB-05	B10E-PS-BOB-05	Dwg 1224211 Sheet 9	Digital		0	1oo7	J	N/a		B10E-SIF-08			CL	CL	CL			CL	CL	CL	CL	CL	CL																									
2.0	BOB06[j]	Beam Off Button 06	B10E-PS-BOB-06	B10E-PS-BOB-06	Dwg 1224211 Sheet 9	Digital		0	1oo7	J	N/a		B10E-SIF-08			CL	CL	CL			CL	CL	CL	CL	CL	CL																									
2.0	BOB07[j]	Beam Off Button 07	B10E-PS-BOB-07	B10E-PS-BOB-07	Dwg 1224211 Sheet 9	Digital		0	1oo7	J	N/a		B10E-SIF-08			CL	CL	CL			CL	CL	CL	CL	CL	CL																									
2.0	BOB08[j]	Beam Off Button 08	B10E-PS-BOB-08	B10E-PS-BOB-08	Dwg 1224211 Sheet 9	Digital		0	1oo7	J	N/a		B10E-SIF-08			CL	CL	CL			CL	CL	CL	CL	CL	CL																									
4.0	RDMND01	Radiation Monitor Accumulated Dose	B10E-PS-RADMN-01	B10E-PS-RDMND-01	Dwg 1224211 Sheet 10	Analogue 4-20mA		>4mA	1oo2	K	N/a		B10E-SIF-09			CL	CL	CL			CL	CL	CL	CL	CL	CL																								CL	
4.0	RDMNR01	Radiation Monitor Dose Rate	B10E-PS-RADMN-01	B10E-PS-RDMNR-01	Dwg 1224211 Sheet 10	Analogue 4-20mA		>4mA	1oo2	K	N/a		B10E-SIF-09			CL	CL	CL			CL	CL	CL	CL	CL	CL																								CL	
4.0	RDMND02	Radiation Monitor Accumulated Dose	B10E-PS-RADMN-02	B10E-PS-RDMND-01	Dwg 1224211 Sheet 10	Analogue 4-20mA		>4mA	1oo2	K	N/a		B10E-SIF-09			CL	CL	CL			CL	CL	CL	CL	CL	CL																								CL	
4.0	RDMNR02	Radiation Monitor Dose Rate	B10E-PS-RADMN-02	B10E-PS-RDMNR-01	Dwg 1224211 Sheet 10	Analogue 4-20mA		>4mA	1oo2	K	N/a		B10E-SIF-09			CL	CL	CL			CL	CL	CL	CL	CL	CL																								CL	
2.0	GADL01	General Access Door Locked 01	B10E-PS-DS-01	B10E-PS-GADL-01	Dwg 1224211 Sheet 8	Digital		0	1oo2	L	N/a		B10E-SIF-07																	On^c									On^c												
2.0	GADL02	General Access Door Locked 02	B10E-PS-DS-02	B10E-PS-GADL-02	Dwg 1224211 Sheet 8	Digital		0	1oo2	L	N/a		B10E-SIF-07																	On^c									On^c												
2.0	GADL03	General Access Gate Locked 03	B10E-PS-DS-04	B10E-PS-GADL-03	Dwg 1224211 Sheet 8	Digital		0	1oo2	M	N/a		B10E-SIF-07																	On^c									On^c												
2.0	GADL04	General Access Gate Locked 04	B10E-PS-DS-05	B10E-PS-GADL-04	Dwg 1224211 Sheet 8	Digital		0	1oo2	M	N/a		B10E-SIF-07																	On^c									On^c												
2.0	SADL01	Service Access Door Locked 01	B10E-PS-DS-07	B10E-PS-SADL-01	Dwg 1224211 Sheet 8	Digital		0	1oo2	N	N/a		B10E-SIF-07																	On^c									On^c												
2.0	SADL02	Service Access Door Locked 02	B10E-PS-DS-08	B10E-PS-SADL-02	Dwg 1224211 Sheet 8	Digital		0	1oo2	N	N/a		B10E-SIF-07																	On^c									On^c												
2.0	ASB01	Area Search Button 01	B10E-PS-ASB-01	B10E-PS-ASB-01	Dwg 1224211 Sheet 5	Digital		N/a	Sequential	N/a	$T3^d$		B10E-SIF-04								CL_{T3}	CL_{T3}	CL_{T3}	CL_{T3}				On_{T3}				On_{T3}	On_{T3}		On_{T3}	On_{T3}	On_{T3}	On_{T3}			On_{T3}	On_{T3}	On_{T3}	On_{T3}	On_{T3}								
2.0	ASB02	Area Search Button 02	B10E-PS-ASB-02	B10E-PS-ASB-02	Dwg 1224211 Sheet 5	Digital		N/a	Sequential	N/a	T4		B10E-SIF-04 & 05								CL_{T4}	CL_{T4}	CL_{T4}	CL_{T4}				On_{T4}				On_{T4}	On_{T4}		On_{T4}	On_{T4}	On_{T4}	On_{T4}			On_{T4}	On_{T4}	On_{T4}	On_{T4}	On_{T4}								
2.0	ASB03	Area Search Button 03	B10E-PS-ASB-03	B10E-PS-ASB-03	Dwg 1224211 Sheet 5	Digital		N/a	Sequential	N/a	T5		B10E-SIF-04 & 05								CL_{T5}	CL_{T5}	CL_{T5}	CL_{T5}				On_{T5}				On_{T5}	On_{T5}		On_{T5}	On_{T5}	On_{T5}	On_{T5}			On_{T5}	On_{T5}	On_{T5}	On_{T5}	On_{T5}								
2.0	ASB04	Area Search Button 04	B10E-PS-ASB-04	B10E-PS-ASB-04	Dwg 1224211 Sheet 5	Digital		N/a	Sequential	N/a	T6		B10E-SIF-04 & 05								CL_{T6}	CL_{T6}	CL_{T6}	CL_{T6}				On_{T6}				On_{T6}	On_{T6}		On_{T6}	On_{T6}	On_{T6}	On_{T6}			On_{T6}	On_{T6}	On_{T6}	On_{T6}	On_{T6}								
2.0	ASBF	Area Search Button Final	B10E-PS-ASBF-01	B10E-PS-ASBF-01	Dwg 1224211 Sheet 7	Digital		N/a	Sequential	N/a	$T7^h$		B10E-SIF-06								CL_{T7}	CL_{T7}	CL_{T7}	CL_{T7}				On_{T7}				On_{T7}	On_{T7}		On_{T7}	On_{T7}	On_{T7}	On_{T7}			On_{T7}	On_{T7}	On_{T7}	On_{T7}	On_{T7}								
2.0	LCRx[i]	Light Curtain (Receiver)[i]	B10E-PS-LCRx[i]	B10E-PS-LCRx[i]	Dwg 1224211 Sheet 5	Digital		0	Sequential	N/a	$T2^i$		B10E-SIF-04								CL_{T2}	CL_{T2}	CL_{T2}	CL_{T2}				On_{T2}				On_{T2}	On_{T2}		On_{T2}	On_{T2}	On_{T2}	On_{T2}			On_{T2}	On_{T2}	On_{T2}	On_{T2}	On_{T2}								
2.0	SCR	Search Card Reader	B10E-PS-SCR	B10E-PS-SCR	Dwg 1224211 Sheet 6	Digital		N/a	Sequential	N/a	T1		B10E-SIF-06												CL_{T1}	CL_{T1}		On_{T1}			On_{T1}	On_{T1}																					
2.0	KEY02	Hutch Enable Key	B10E-PS-KEY-02	B10E-PS-KEY-02	Dwg 1224211 Sheet 6	Digital		0	1oo1	N/a	T1		B10E-SIF-06												CL_{T1}	CL_{T1}		On_{T1}			On_{T1}	On_{T1}																					
2.0	KEY01[a]	Electron Source Enable Key (ZCP)	B10E-PS-KEY-01	B10E-PS-KEY-01	Dwg 1224211 Sheet 06	Digital		0	1oo1	N/a	T1		B10E-SIF-10												CL_{T1}	CL_{T1}		On_{T1}			On_{T1}	On_{T1}																					
2.0	KEY03[a]	Electron Source Enable Key (Rack)	B10E-PS-KEY-03	B10E-PS-KEY-03	Dwg 1224211 Sheet 10	Digital		0	1oo1	N/a	N/a		B10E-SIF-10						OP	OP																															